

Raza Shaud

Roll No.: 264102509
M.Tech — Electronics and Electrical Engineering
Specialization: Systems, Control and Automation
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (EEE)	Indian Institute of Technology, Guwahati	Pursuing	2026–2028
B.Tech. (Electrical Engineering)	Ghani Khan Choudhury Institute of Engg. and Tech., Malda	7.54	2022–2026
Senior Secondary	WBCHSE	71.80%	2022
Secondary	WBBSE	65.40%	2020

EXPERIENCE

- Tech Solutions Inc.** May 2025 – Jul. 2025
Machine Learning Intern Kolkata, India
 - Optimized **8** machine learning scripts, improving model prediction speed by **30%**.
 - Tested and validated classification models with a 4-person team.
- Steel Authority of India Limited (SAIL)** Nov. 2024 – Dec. 2024
Internship Trainee — Industrial Automation & Plant Electrical Systems Bokaro, India
 - Studied plant electrical systems, protection schemes and automation processes across an integrated steel works.
- Damodar Valley Corporation (DVC)** Jun. 2024 – Jul. 2024
Internship Trainee — Thermal Power Generation, Switchgear & Protection Mejia, India
 - Documented daily operations, switchgear arrangements and protection settings on an operating thermal generating station.

PROJECTS

- Solar-Powered Robotic Cable Monitoring System** Aug. 2025 – Apr. 2026
B.Tech Major Project | Arduino, Yolo v8, CNN, OpenCV, Solar PV Github
 - Built an autonomous inspection robot on an **Arduino Uno** with **12V/20W solar PV** supply and camera-based insulation monitoring over **500 m** of cable.
 - Trained a **Yolo** fault classifier on **1,300+** cable images reaching **96.5% accuracy** and **78%** fewer false positives; tuned the power budget for **8 h** of continuous field operation.
- Saline Level Monitoring and Alert System (IoT)** Jan. 2025 – Apr. 2025
Self Project | Arduino Uno, Ultrasonic Sensor, GSM Module Github
 - Interfaced an ultrasonic sensor with an **Arduino** microcontroller in **Embedded C** to track fluid level to **2 cm** precision; validated over **50 hours** with zero missed empty-bottle events.
 - Implemented a dual alert path (buzzer + **GSM SMS** over UART to 3 contacts), reducing simulated response time by **75%** and nursing workload by **40%**.
- Human vs. AI Voice Classification System** Sep. 2024 – Nov. 2024
Self Project | Python, Scikit-learn, Digital Signal Processing Github
 - Extracted **MFCC** features from **1,500+** audio samples with spectral-gating noise reduction (**65%** less interference) to build a binary classifier at **89% accuracy**.
 - Deployed a **Random Forest** on a 15-D feature vector, beating an SVM baseline by **12%** via 5-fold cross-validation.

TECHNICAL SKILLS

- Programming:** C, C++, Python
- Embedded & Hardware:** Arduino Uno, Sensor Interfacing, Hardware Prototyping & Debugging
- Signal Processing & ML:** DSP, Feature Extraction, Yolo v8, CNN, OpenCV, Scikit-learn, NumPy, Pandas
- EDA & Tools:** MATLAB/Simulink, ANSYS, ETAP, Git, Arduino IDE, Linux*, $\LaTeX$ * Elementary proficiency

PUBLICATIONS

- IEEE GIST 2024, Best Paper Award:** Dual Rotor Single Stator 6-Phase PMSM for Smart Electric Vehicles [Link]
- Tech Research Chronicles (IAST), 2025:** Noise Filtration and Voice Discrimination: Case Studies and Results [Link]
- PEIS-2026, GKCIET Malda:** Intelligent Remote Monitoring of Low Voltage Cable Insulation Fault [Link]

POSITIONS OF RESPONSIBILITY

- Class Representative (CR),** Dept. of Electrical Engineering, GKCIET Malda 2024 – 2026
- Organizing Committee Member,** PEIS-2026 International Conference, GKCIET Malda Apr. 2026

ACHIEVEMENTS

- GATE 2026 (Electrical Engineering):** All India Rank **2139**, admitted to M.Tech at IIT Guwahati 2026
- 100 Days of Code: Complete Python Pro Bootcamp,** Udemy Certification 2024